

Game for Government and Forestry Enterprise of the Supervision and Management Behaviors

Sun Wenqi

College of Economics and Management
Northeast Forestry University
Harbin, China
Sunwenqi800@yahoo.com.cn

Cao Yukun

College of Economics and Management
Northeast Forestry University
Harbin, China
cyklk@163.com

Abstract—At present, the contradictory conflict exists between government and forestry enterprise in forestry management objectives. This paper studies the effect of the supervision cost and punishment on the game behaviors between government and forestry enterprise. Four types of game were studied in this paper, including complete information static game, complete information dynamic game, incomplete information static game and incomplete information dynamic game. The results indicated that the Nash equilibrium of complete information static game is related to the income and punishment of forestry enterprise to take illegal management and the loss and supervising cost of government. Under complete information dynamic condition, the game player who firstly takes strategies has disadvantage. The game result of incomplete information static game is same to that of complete information static game. Under incomplete information dynamic condition, forestry enterprise trends to take illegal strategies. This paper proposed that government should construct the encouraging and supervising mechanisms and further perfect the forestry cutting rights, forestry resources information and supervising systems.

Index Terms—forestry right reform, game, equilibrium, government, forestry enterprise, supervision

I. INTRODUCTION

In the nineteen eighties, northeast state-owned forest region began to explore the forestry right system reform. Yichun forest region firstly has carried out various pilot activities, including afforestation on their own expense, construction of commercial forestries and development of private forestry. The development of Yichun promoted the reform of state-owned forest right^[1-5]. However, due to the long-term bondage of planned economy concept and system, the progress of state-owned forest right reform has been slow. Forestry development goal of government is the maximization of forestry ecological, economical and social benefits. At present, government's goal was to achieve ecology benefit. However, the forestry management goal of forestry enterprise and worker is to achieve economic benefits. This leads to the contradictory conflict of government and forestry enterprise in forestry management objectives. Forestry enterprise builds ten thousand mu of forest in the process of building in forestry. Because a series of limited system such as cutting quota system, they get the lower economic benefits. Moreover, they often have liabilities, and can not make the resource advantages into economic benefits.

After forestry right reform, the traditional forestry enterprise were changed to modern forestry enterprise. The main difference of modern and traditional forestry enterprise is separation of the ownership and management. Government does not directly dispose capital, but entrusted forestry enterprise to implement management and decision-making, forming the relationship of entrusting-agent. Because the interests of government departments, enthusiasm of forestry enterprise for production is not high. Thus, responsibility to protect forestry resources can not be implemented by forestry enterprise. The sustainable development of forestry faced serious challenges and forestry resources increased slowly. At present, the state-owned forest right reform is at the situation of bottleneck. In this management process, the objective functions of forestry enterprise and government are not identical, so there is a conflict of interest. Thus, forestry enterprise often takes illegal management. The way to solve this problem is that government implements an effective supervision system on forestry enterprise^[6-8].

This paper studies the effect of the supervision cost on the game behaviors of government and forestry enterprise. Four types of game were studied in this paper, including complete information static game, complete information dynamic game, incomplete information static game and incomplete information dynamic game.

II. BASIC ASSUMPTIONS AND MODEL CONSTITUTION

A. Complete Information Static Game

For forestry enterprise, the assumed choice of pure strategies is legal and illegal managements. Government has two options of supervision and without supervision. Whether government takes supervision, additional income of forestry enterprise is 0 when his management is legal. K represents the additional income of forestry enterprise when his management is illegal. When forestry enterprise's illegal management occurs, government's loss may not only K , but also ecological, social and environmental losses. So R represents government's loss when government does not supervise. The R and K are related. Clearly, K becomes larger, R becomes larger. C represents supervision cost of government and F represents illegal management cost (when government found illegal management, government will take penalties to enterprise). If government

implements supervision, forestry enterprise's illegal management will be found, and the $C < F$ occurs here. Since the game is a complete information static game, the information is common knowledge. Table 1 corresponds to the different combistates of pure strategies in payoff matrix. Now, we solve the mixed strategies Nash equilibrium.

TABLE I. COMPLETE INFORMATION STATIC GAME CORRESPONDING TO DIFFERENT COMBISTATES OF PURE STRATEGIES IN PAYOFF MATRIX

		Forestry enterprise	
		Illegal management	Legal management
Government	Supervision	$F-C, -F$	$-C, 0$
	Without Supervision	$-R, K$	$0, 0$

The probabilities for government to take supervision and without supervision are p and $(1-p)$, respectively. The probabilities for forestry enterprise to choose illegal and legal managements are q and $(1-q)$. As q was determined, the expected income of government for choosing supervision is shown (1):

$$Y0(\text{Supervision}, q) = (F-C)q + (-C)(1-q) \quad (1)$$

Namely,

$$Y0(\text{Supervision}, q) = Fq - C \quad (2)$$

The expected income of government for choosing without supervision is shown (3):

$$Y0(\text{Without supervision}, q) = (-R)q + 0(1-q) \quad (3)$$

Namely,

$$Y0(\text{Without supervision}, q) = -Rq \quad (4)$$

Based on (2) and (4), the (5) was obtained as shown following.

$$Y0(\text{Supervision}, q) = Y0(\text{Without supervision}, q) \quad (5)$$

Through calculating the (5), the solution of (5) was shown in (6)

$$q^* = C/(F+R) \quad (6)$$

As p was determined, the expected income of forestry enterprise for choosing illegal management is shown (7):

$$Y1(p, \text{Illegal management}) = (-F)p + K(1-p) \quad (7)$$

Namely:

$$Y1(p, \text{Illegal management}) = K - (F+K)p \quad (8)$$

The expected income of forestry enterprise for choosing legal management is shown (9):

$$Y1(p, \text{Legal management}) = 0 \quad (9)$$

Based on (8) and (9), the (10) was obtained as shown following.

$$Y1(p, \text{Illegal management}) = Y1(p, \text{Legal management}) \quad (10)$$

Through calculating the (10), the solution of (10) was shown in (11)

$$p^* = K/(F+K) \quad (11)$$

For forestry enterprise, if the probability of government to implement supervision is less than p^* , the optimal choice of forestry enterprise is taking illegal management. If the probability of government for implementing supervision is more than p^* , the optimal choice of forestry enterprise is taking legal management. If the probability of government for implementing supervision is equal to p^* , forestry enterprise does randomly choose legal or illegal managements.

For government, if the probability of forestry enterprise for taking illegal management is greater than q^* , government's best option is supervision. If the probability of forestry enterprise for taking illegal management is less than q^* , government's optimal choice is without supervision. If the probability of forestry enterprise for taking illegal management is equal to q^* , government can randomly select supervision and without supervision.

Therefore, the mixed strategies Nash equilibrium is $p^* = K/(F+K)$ and $q^* = C/(F+R)$. Namely, the probability of government to choose supervision is $p^* = K/(F+K)$, the probability of forestry enterprise to choose illegal management is $q^* = C/(F+R)$.

The Nash equilibrium of this game is related to the additional income K , government loss R , and illegal management cost F and supervising cost C . When F and R are higher, the probability of forestry enterprise taking illegal management is smaller. And C increased, the probability of forestry enterprise for taking illegal management is larger. Why the probability of forestry enterprise for choosing illegal management increased with increasing K ? Because K increased, the loss of government becomes larger. Namely, R value increased, thus the probability for government to implement supervision increased. Compared to K , the change in C is very slight. Therefore, when K increased, the risks for forestry enterprise to take illegal management increased. At this time, the probability of government implementing supervision also increased. Thus, the illegal behaviors of forestry enterprise are found relative easily. Then, forestry enterprise does not dare to choose illegal behavior. It can be seen that the optimal probability p^* of government for implementing supervision determined the income maximum of forestry enterprise. The p^* is dependent on additional income of K and illegal management cost of F .

B. Complete Information Dynamic Game

We discussed two cases of government to act first and forestry enterprise to act first, respectively.

1) Forestry enterprise first taking action.

If forestry enterprise takes illegal management strategies (see Table 1) and government implements supervision due to both sides information being complete, forestry enterprise's illegal management will be found. Thus, forestry enterprise does not take the bad strategies of illegal management. If forestry enterprise takes legal management strategies and government takes the strategies of without supervision, thus

two sides have no loss of any cost. Then, the Nash equilibrium solution for the game is (without supervision, legal management).

2) Government first taking action.

Forestry enterprise takes strategy after observing government's actions, and the game is limited (see Table 1). If government chooses to supervise, the optimal strategy of forestry enterprise is to choose legal management after observing government's actions. If forestry enterprise takes illegal management, he will lose the cost of F due to the punishment of government. If government does not supervise, the optimal choice is illegal management when forestry enterprise observed government's actions. In this case, forestry enterprise will get more income because of $K > 0$. Thus, according to government's strategies, we can get the Nash equilibrium of complete information dynamic game, that government and forestry enterprise adopt the strategies of (supervision, legal management) and (without supervision, illegal management).

After one side taking action, the other side will make the appropriate strategies according to the corresponding situation. Namely, there is disadvantage for the first acted side. Forestry enterprise who has potential violation motivation hopes government first to take action in order to better implements their own illegal management.

3) Incomplete information static game

Although government and forestry enterprise have shared information, in actual operation, some information on the two sides have been reserved. If government found illegal management of forestry enterprise, government's utility function is $(F-C+t_1)$, where t_1 is the income corresponding to government's private information.

If government does not supervise and forestry enterprise takes legal management, the forestry enterprise's utility is $(0+t_2)$, where t_2 is an income corresponding to forestry enterprise's private information. The t_1 and t_2 here are the uniform distribution of the two independent variables from $[0, x]$. The t_1 is only known by government and t_2 is only known by forestry enterprise, forming a static Bayesian payoff matrix as shown in Table 2.

TABLE II. INCOMPLETE INFORMATION STATIC GAME CORRESPONDING TO DIFFERENT COMBISTATES OF PURE STRATEGIES PAYOFF MATRIX

		Forestry enterprise	
		Illegal management	Legal management
Government	Supervision	$F-C+t_1, -F$	$-C, 0$
	Without supervision	$-R, K$	$0, 0+t_2$

The following pure strategies were considered. There is a $t_1^* \in [0, x]$ and a $t_2^* \in [0, x]$. When $t_1 \geq t_1^*$, government takes supervision. When $t_1 < t_1^*$, government does not take supervision. If $t_2 < t_2^*$, forestry enterprise will choose illegal management and if $t_2 \geq t_2^*$, forestry enterprise will take legal management. Therefore, the probability of government to choose supervision is $1-t_1^*/x$ and that not to choose

supervision is t_1^*/x . The probability of forestry enterprise to choose legal management is t_2^*/x and that to choose illegal management is $1-t_2^*/x$.

As the strategies of government were determined, the expected utility of forestry enterprise to choose illegal management was shown in (12):

$$(-F)(1-t_1^*/x) + Kt_1^*/x = (F+K)t_1^*/x - F \quad (12)$$

The expected utility of forestry enterprise to choose legal management was shown in (13):

$$0(1-t_1^*/x) + (0+t_2)t_1^*/x = t_1^*t_2^*/x \quad (13)$$

The t_2^* should satisfy with (14):

$$(F+K)t_1^*/x - F = t_1^*t_2^*/x \quad (14)$$

Same, as the strategies of forestry enterprise were determined, expected utility of government to take supervision is shown in (15)

$$(F-C+t_1)(1-t_2^*/x) + (-C)t_2^*/x = (F+t_1)(1-t_2^*/x) - C \quad (15)$$

Expected utility of government to take without supervision is shown in (16)

$$(-R)(1-t_2^*/x) + 0t_2^*/x = Rt_2^*/x - R \quad (16)$$

The t_1^* should satisfy with (17):

$$(F+t_1^*)(1-t_2^*/x) - C = Rt_2^*/x - R \quad (17)$$

Based on (14) and (17), the solutions were obtained as shown in (18) and (19).

$$t_1^* = [(EG+N) + ((EG+N)^2 - 4WEJ)^{1/2}] / (2W) \quad (18)$$

$$t_2^* = G - 2JW / [(EG+N) + ((EG+N)^2 - 4WEJ)^{1/2}] \quad (19)$$

Where $W = 1 - (F+K)/x$; $E = (R+F)/x$; $N = C - 2F - R$; $G = F+K$; $J = Fx$

Therefore Bayesian equilibrium is obtained: (1) government: when $t_1 \geq t_1^*$, government takes supervision and when $t_1 < t_1^*$, government does not take supervision. (2) Forestry enterprise: when $t_2 \geq t_2^*$, forestry enterprise takes illegal management and when $t_2 < t_2^*$, forestry enterprise takes legal management. Therefore, forestry enterprise thinks that the probability of government for choosing supervision is $1-t_1^*/x$, government considers that the probability of forestry enterprise to choose illegal management is $1-t_2^*/x$. The probability not to take supervision is t_1^*/x and the limitation of t_1^* when x tends to 0 was shown (20):

$$\lim_{x \rightarrow 0} t_1^* = \lim_{x \rightarrow 0} [(EG+N) + ((EG+N)^2 - 4WEJ)^{1/2}] / (2W) = 1 - K / (F+K) \quad (20)$$

The probability of government to supervise is $K/(F+K)$;

Same, the limitation of t_2^* when x tends to 0 is as shown in (21)

$$\lim_{x \rightarrow 0} t_2^* = \lim_{x \rightarrow 0} G - 2JW / [(EG+N) + ((EG+N)^2 - 4WEJ)^{1/2}] = 1 - C / (F+R) \quad (21)$$

The probability of forestry enterprise to take illegal management is $C/(F+R)$.

These results indicated that the results of incomplete information static game is same to the game result of complete information static game.

4) Dynamic games with incomplete information

First, we remove the bad strategies, which can reduce the number of game balance, simplify the game process. In table 2, for each information set, when the faith of player to the information set and the strategies of other players were determined, the acted player should take the best strategies. When the belief of forestry enterprise in imperfect information set was given, the expected utility of forestry enterprise to adopt illegal management is shown in (22):

$$K-(F+K)p \quad (22)$$

If enterprise do not take illegal management, the corresponding expected utility is $0P+(1-P)t_2 \approx 0$ (t_2 is a small positive number). The calculated result was shown in (23):

$$[K-(F+K)p]-0=K-(F+K)p \quad (23)$$

K is the illegal income of forestry enterprise and F is cost for illegal management due to the punishment of government. We believe that the value of $K-(F+K)p$ tends to be greater than 0 due to $0 \leq p \leq 1$ based on (23). Therefore, forestry enterprise will not adopt legal management. Whether or not government takes supervision, forestry enterprise should take illegal management. This is his best response to government's strategies. Thus, given the posterior probability and strategy of forestry enterprise, the best strategy is illegal management. So (supervision, illegal management) is a refined Bayesian equilibrium.

III. EFFECTIVE POLICIES OF GOVERNMENT FOR IMPLEMENT SUPERVISION

A. Construction of the Supervision Mechanisms of Interests

The supervision mechanism of interest means that the behaviors of people to obtain benefit are limited by the law and morality. Law and morality are the regulators and controllers of the interest's needs and behaviors. By strengthening the law construction, we strengthen the law constraints to managers. State guides people to obtain benefits through legal methods, preventing illegal methods such as trading by exchange of power and money, as well as to reduce social contradictions and interest conflicts. Morality is an intrinsic binding force, which can lead people to determine a reasonable target of interest and to select the reasonable interest behavior.

B. Improving the Supervision Mechanism of State-owned Forest Resources

With the deepening of reform on forest resources management system, the community proposes new requirements to strengthen the supervision of forestry resources. According to studies on the game of interests, we also get this conclusion that under conditions of complete information, choosing the optimal strategy of forestry enterprise will depend

upon the probability of government to take supervision strategy. The reform of the supervision system for forestry resources also should be on the agenda. Increasing supervision strength of forestry resources and constructing of an effective supervision mechanism for forestry resources are to guarantee successful reform of state-owned forest.

The production supervision mechanism of the forestry resources of state-owned forest should change to ecological supervision mechanism. In the key state forestry, in the process of state to implement the resource supervision of forest, state need to take innovative concepts, where the state understand problem on the supervision of forest from the perspective of ecological construction. Meanwhile, the state should put other resources interperedly related to forest resources into the scope of supervision.

First, government can learn from foreign advanced forestry practices in developed countries. Government can change the past resource supervision system to vertical management. Therefore, government can establish authority and effective management department, which represents the state to exercise the ownership, management, supervision and administrative punishment on the forestry enterprise under its jurisdiction.

Secondly, the functions of supervision system need to be determined clearly. On the one hand, we must be clear that the main function of the management department is supervision. The other hand, the state should grant to the supervision department for certain hard constraints means within the scope of administrative permission, changing the current status of weak constraints.

Finally, the state should establish the guidance system of personnel fixed configuration corresponding to the supervision management department. It was suggested that the state forestry administration and personnel departments work together, to establish a national supervisory authority (supervision department) and guidance system of personnel fixed configuration, which can provide policy basis for the local forestry authorities to seek agency staff, and also create the policy conditions for the progress of the supervision system.

C. Improving the Supervision Mechanism of Forestry Harvesting Rights

Cutting management of forestry is an important institutional guarantee for the sustainable development of forestry and forestry ecosystem conservation. It is also an important institutional guarantee for the balance between the public ecological security interests and private economic interests. A certain extent, the present harvesting management system led to that rights and interests can not be achieved by forestry enterprise and forestry workers in the contract. This system reduces the investor enthusiasm for forestry management, and in some degree it restricts the further development of non-public forestry.

According to the reform of state-owned forest, the present reform ideas for cutting right are that the fundament is classification management of forestry. And state should strictly

manage the public forestry. At the same time, state needs to liberalize the commercial forestry management. The reformed system should guarantee that the approval process be simplified. And approval content is simple and cutting plan allocation is fair and transparent. After reform, it was truly realized that owners who have the ownership of forestry resources can take fairly quick cutting index. And disposition right of forestry harvesting was implemented for forestry enterprise. In addition, commercial forestry management rights are flexible for forestry enterprise. Thus, enthusiasm of forestry enterprise for afforestation and forestry protection is activated maximally, which promotes the rapid development of forestry.

D. Building Mechanisms for Binding and Stimulating of Forestry Enterprise

Based on the game results of forestry enterprise and government on the supervision and management, the incentive and restrictive mechanisms were affected by the probability of government for choosing supervision and the income of forestry enterprise.

The reason of forestry enterprise to choose illegal management strategy is due to information asymmetry and opportunistic behavior of forestry enterprise. Forestry enterprise has a lot of assets in production, sales areas and service areas and so on. Operators can use their power to make profits and harm the interests of government. Although forestry enterprise has motivation of illegal management, but only if certain constraints change, such as institutional constraints, forestry enterprise will change from the potential violators into reality offenders to seek their maximum gain. However, the motivation for illegal management of forestry enterprise does not constitute a sufficient condition for illegal management. According to the cost-benefit analysis for forestry enterprise taking legal and illegal managements, government may try to establish an incentive system and restriction mechanisms in forestry enterprise. Such a mechanism could increase the illegal management cost of forestry enterprise of F and reduce the illegal income of R , which makes the illegal income $R < \text{illegal management cost } F$. Therefore, the illegal behaviors of forestry enterprise will change into legal behaviors.

E. Improving the Information System of State-owned Forest Region

The construction of forestry information system of state-owned forest areas is through using network resources to build forestry network management system. Thus, the state and local forestry administration can share the information resources to achieve intersectional collaboration working, instead of the traditional management mechanism. The construction of state forestry information system should be uniformly designed, and

implemented step by step under the guidance of state according to the local conditions. This system should define the various stages of development objectives and main content. It was needed to break the information blockade and sectoral fragmentation concept. The system provides the basis information to the industry and other industries for forestry construction, including forestry resources and personnel archives, realizing resource and information sharing.

With the deepening of state-owned forest right reform, the government should strengthen the investigation work of the plan and design of non-public forest, and build a professional non-public forestry archives. State should know the status of non-public forestry resources. And state should assess the effectiveness of forestry management. Thus, state can provide reliable evidence to the preparation of forestry design, determine the forestry management measures.

Forestry departments should set up a professional archives management department for non-public forest. Government should guide and strengthen the management of the archives management department, to ensure the authenticity, science and continuity of the non-public forest archives.

ACKNOWLEDGMENT

The author would like to acknowledge the Fund Project: Ministry of Education, Humanities and Social Sciences Planning Fund (08JA630014)

REFERENCES

- [1] Y. H. Wang, "Some issues about the reform and development in northeast state-owned forest areas," *Forestry Economics. China*, vol. 1, pp. 22-26, January 2006.
- [2] G. C. Liu, Y. X. Ren, and Z. Bao, "Probe of the reform of forestry property system in northeast state-owned forest region," *China Forestry Economy. China*, vol. 1, pp. 7-10, January 2007.
- [3] S. J. Liu, and J. W. Hua, "Investigation and thinking of Yichun forestry tenure reform," *Forestry Economics. China*, vol. 9, pp. 48-52, July 2007.
- [4] J. F. Lei, "Some thoughts on reform of chinese key state owned forestry regions," *Journal of Beijing Forestry University. China*, vol. 28, pp. 143-146, September 2006.
- [5] Z. Li, "Reflections on the reform of state-owned forest-policy suggestion to natural fatural forestry protection project in the state areas," *Forestry Economics. China*, vol. 5, pp. 42-47, 1999.
- [6] L. Gao, D. S. Liu, and J. Lu, "State-owned forest resources monitoring system and mechanisms in northeast and inner mongolia forestry areas: problems and causes analysis, forestry economics," *China*, Vol. 2, pp. 87-91, July 2010.
- [7] D. S. Liu, and J. Lu, "Study on the reform of chinese forestry resources supervision system, problems of forestry economics," *China*, Vol. 27, pp.489-493, octouber 2007.
- [8] M. Y. Jiang, and J. P. He, "Thinking of the construction of supervision system of forestry resource in state -owned forestry region," *China Forestry Economy, China*, Vol. 6 pp. 8-10, 2006